

KNOWLEDGE MANAGEMENT

UNIT I — BUSINESS INTELLIGENCE AND BUSINESS DECISIONS

Detailed, corrected, easy-to-learn and examination-focused notes

Course: BCA Semester V | **Course Code:** BCA-5001 | **Unit:** I | **Focus:** BI, DSS, GDSS, groupware, data warehousing and decision-making

Study method: Read the concept, learn the exam answer, revise the quick sheet and practise the questions.

1. Unit-I Syllabus Map

Topic	Coverage
Modeling decision process	Decision steps, rationality, bounded rationality, intuitive and creative models
Decision Support Systems	Meaning, components, types, characteristics and applications
Group Decision Support Systems	Meaning, components, phases, features, advantages and limitations
Groupware technologies	Synchronous/asynchronous groupware, tools and applications
Business Intelligence	Goals, components, process, tools, analytics and applications
OLTP, OLAP, data warehouse and data marts	Definitions, differences and architecture
Data mining and knowledge discovery	Meaning, process and techniques

2. Learning Objectives

After completing this unit, you should be able to:

- Define BI, DSS, GDSS and groupware in exam-ready language.
- Explain how raw data becomes information, insight and action.
- Describe components, types, advantages and limitations of DSS and GDSS.
- Differentiate DSS from GDSS and a conventional Information System.
- Explain synchronous and asynchronous groupware with examples.
- Explain OLTP, OLAP, data warehouses, data marts, data mining and KDD.

3. Business Intelligence

3.1 Meaning and Definition

Business Intelligence (BI) is a combination of methods, processes, architectures and technologies used to collect, integrate, store, analyse and present business data in a meaningful form. Its purpose is to support better managerial decisions and improve organizational performance.

In simple words, BI converts **raw data** into **information**, information into **insight**, and insight into **action**. For example, a retailer can analyse sales records to discover which products sell most, in which season, in which region and to which customers.

Exam definition: Business Intelligence is a set of technologies and analytical processes that transform organizational data into useful information and insights for strategic, tactical and operational decision-making.

3.2 Goals of BI

- Improve decision-making with accurate, timely and relevant information.
- Identify business opportunities through trends, patterns and customer behaviour.
- Improve efficiency by locating waste, bottlenecks and unnecessary costs.

- Create competitive advantage through market and competitor analysis.
- Build a data-driven organizational culture.
- Improve customer satisfaction by understanding needs and complaints.
- Monitor KPIs through dashboards and alerts.

3.3 Components of BI

Component	Explanation	Example
Data sources	Internal and external systems that generate data.	ERP, CRM, transactions, spreadsheets, APIs, IoT
Data warehouse	Central repository of integrated historical data.	Multi-branch sales warehouse
ETL / ELT	Extracts, cleans, transforms and loads data.	Combining online and offline purchases
Business analytics	Uses queries, statistics, models and data mining.	Sales forecasting
Reporting and visualization	Communicates findings through reports and dashboards.	Regional sales dashboard
Performance management	Connects KPIs with business objectives.	Tracking revenue and retention

3.4 BI Process: Data to Decision

1. Define: Identify the business question. **2. Collect:** Gather data from databases, ERP, CRM, websites, spreadsheets, cloud systems, IoT devices and APIs. **3. Prepare:** Remove duplicates, handle missing values and standardize formats. **4. Store:** Place prepared data in a warehouse or data lake. **5. Analyse:** Apply queries, statistics, OLAP, visualization and data mining. **6. Communicate:** Present results with charts, dashboards, reports and alerts. **7. Act and monitor:** Implement decisions, track KPIs and learn from feedback.

BI chain: Data → Integration → Storage → Analysis → Visualization → Insight → Decision → Action → Feedback

3.5 Types of Analytics

Type	Question	Example
Descriptive	What happened?	Monthly sales report
Diagnostic	Why did it happen?	Sales fell due to stockouts
Predictive	What may happen?	Next-month demand forecast
Prescriptive	What should be done?	Recommended stock level

3.6 Common BI Tools

Tool	Main use
Microsoft Power BI	Dashboards, reports, data models and sharing
Tableau	Interactive visualization and visual analytics
Qlik Sense	Associative analytics and interactive reporting
IBM Cognos	Enterprise reporting and performance management
SAP BusinessObjects / Oracle BI	Enterprise reporting and analytics suites
Looker / Spotfire / SAS BI	Data exploration and analytical reporting

3.7 Applications of BI

- Sales and marketing: segmentation, campaign analysis and forecasting.
- Finance and risk: expense analysis, fraud detection and profitability.
- Supply chain: demand forecasting, inventory and logistics optimization.
- Healthcare: patient-flow analysis and resource planning.
- Human resources: recruitment, performance and retention analysis.

- Customer service: complaints, response time and service quality.
- Manufacturing: quality control, production monitoring and predictive maintenance.
- Education: student performance, attendance and personalized learning.

3.8 Advantages and Limitations

Advantages: improved decisions, faster reporting, better data access, higher productivity, improved data quality, lower costs, stronger customer understanding and competitive advantage.

Limitations: implementation cost, data-quality dependence, privacy and security risks, integration complexity, need for trained staff, analytical bias and overdependence on dashboards.

Exam Answer: Business Intelligence

Question: What is Business Intelligence? Explain its components and advantages.

Answer: Business Intelligence transforms raw organizational data into meaningful information and actionable insights. Its components include data sources, ETL, data warehouse, analytics, reporting and performance management. BI improves data access, decision quality, productivity and cost control, and helps organizations identify opportunities, understand customers and compete effectively.

4. Decision-Making Process

4.1 Meaning

Decision-making is the systematic process of identifying a problem or opportunity, developing alternatives, evaluating them and selecting and implementing an appropriate course of action. Organizational decisions may be strategic, tactical or operational.

4.2 Steps

Step	Explanation
1. Identify	Recognize that a problem, opportunity or improvement need exists.
2. Define	Clarify the problem, causes, objectives, constraints and scope.
3. Gather information	Collect facts, data, rules, expert opinions and historical records.
4. Generate alternatives	Develop possible solutions through research, consultation or brainstorming.
5. Establish criteria	Choose cost, risk, quality, time and feasibility criteria.
6. Evaluate	Analyse benefits, limitations, resources, risks and outcomes.
7. Select	Choose the most suitable alternative using analysis, judgment or consensus.
8. Implement	Prepare an action plan, allocate resources and assign responsibility.
9. Monitor	Measure outcomes against objectives and KPIs.
10. Learn and adapt	Use feedback to correct the decision and improve future decisions.

4.3 Decision-Making Models

Model	Core idea	Example
Rational	Systematic comparison using clear objectives and information.	Supplier scoring
Bounded rationality	Limited time and information lead to a satisfactory option.	First qualified applicant
Intuitive	Uses experience, judgment and pattern recognition.	Experienced doctor
Creative	Generates an original solution.	New business model
Vroom–Yetton	Selects team participation according to the situation.	Manager consults team

5. Decision Support System (DSS)

5.1 Definition

A Decision Support System is an interactive computer-based information system that combines data, analytical models and user-friendly software to help individuals make decisions, especially in semi-structured and unstructured situations. A DSS supports human judgment; it does not replace the manager.

5.2 Components

Component	Function
Data management	Stores and retrieves internal and external data.
Model management	Contains statistical, financial, forecasting, simulation and optimization models.
Knowledge base	Stores rules, expertise and domain knowledge.
User interface	Allows users to enter queries, choose models and view results.
Users	Interpret results, apply judgment and make the final decision.

5.3 Types of DSS

Type	Description	Example
Data-driven	Analyses large datasets and reports.	Sales analysis
Model-driven	Uses mathematical, statistical or optimization models.	Production planning
Communication-driven	Supports teamwork and communication.	Online meeting
Document-driven	Searches documents, emails and web pages.	Legal research
Knowledge-driven	Uses rules, expertise or AI recommendations.	Medical support

5.4 Characteristics

- Interactive and user-controlled.
- Supports, rather than replaces, decision-makers.
- Task-oriented and useful for analysis and alternatives.
- Flexible for regular and ad hoc decisions.
- Combines data with analytical models.
- Improves accuracy, timeliness, quality and effectiveness.

6. Group Decision Support System (GDSS)

6.1 Definition

A Group Decision Support System is an interactive, computer-based system designed to help a group of decision-makers communicate, generate ideas, analyse alternatives, vote and reach a collective decision about a semi-structured or unstructured problem.

DSS generally supports an individual decision-maker, whereas GDSS supports a team or group. GDSS can be used in an electronic meeting room or through online collaboration systems.

6.2 Components

Component	Details
Hardware	Computers, network equipment, displays, projectors, microphones, cameras and meeting facilities.
Software tools	Questionnaires, brainstorming, idea organization, ranking, voting and documentation.
People	Participants, facilitator, decision-makers, technical staff and experts.
Network	Connects participants, facilitator systems, file servers and shared workspaces.

6.3 Meeting Phases

- Problem definition and agenda setting.
- Idea generation and anonymous input.

- Sharing and organizing ideas.
- Discussion and clarification.
- Evaluation and prioritization.
- Voting, ranking or consensus building.
- Recording the decision and action plan.

6.4 Features

- Supports simultaneous input from multiple participants.
- Encourages participation and reduces domination by one person.
- May provide anonymous contribution for honest opinions.
- Supports idea generation, discussion, voting and documentation.
- Allows local and geographically distributed participation.
- Supports intelligence, design, choice and implementation phases.

6.5 Advantages and Limitations

Advantages	Limitations
Better decisions through diverse viewpoints.	Training and learning curve may be needed.
Higher participation and inclusiveness.	Depends on technology and connectivity.
Time and travel savings.	Implementation and maintenance may be costly.
Structured and documented meetings.	Security and privacy concerns.
Remote collaboration and anonymous input.	Groupthink or technology overdependence may occur.

DSS versus GDSS

Basis	DSS	GDSS
Primary user	Individual manager	Group or team
Main focus	Individual analysis	Collaboration and collective decision
Special tools	Models and what-if analysis	Brainstorming, voting and shared workspace
Communication	Not central	Essential
Example	Sales forecasting system	Vendor-selection meeting

7. Groupware Technologies

7.1 Meaning

Groupware is software that enables people to communicate, coordinate and collaborate over a network. It supports shared work even when team members are in different locations or work at different times.

7.2 Synchronous and Asynchronous Groupware

Type	Meaning	Examples
Synchronous	Participants collaborate at the same time in real time.	Chat, video/audio conference, whiteboard, screen sharing
Asynchronous	Participants contribute at different times.	Email, forums, workflow, file sharing, documents, calendars

7.3 Tools and Applications

- Email and discussion forums for archived communication.
- Chat and instant messaging for real-time text.

- Video and audio conferencing for distributed meetings.
- Screen sharing for training and troubleshooting.
- Electronic whiteboards for shared drawing and annotation.
- Shared documents for co-authoring.
- Shared calendars for scheduling.
- Workflow and project-management tools for tasks and approvals.
- Applications include online education, software teams, remote training, customer support and management meetings.

8. OLTP, OLAP, Data Warehouse and Data Mart

8.1 OLTP

Online Transaction Processing supports routine daily transactions such as order entry, payment, ticket booking and attendance recording. It is optimized for fast insert, update and delete operations and current detailed data.

8.2 OLAP

Online Analytical Processing supports multidimensional analysis of historical and integrated data. Its operations include roll-up, drill-down, slice, dice and pivot across time, product, region and customer.

8.3 Data Warehouse and Data Mart

A data warehouse is a centralized repository of integrated historical data designed for analysis and reporting. A data mart is a smaller subject-specific part of a warehouse, such as a sales, finance or HR data mart.

Basis	OLTP	OLAP / Warehouse
Purpose	Daily transactions	Analysis and decision support
Data	Current and detailed	Historical and integrated
Operations	Insert, update, delete	Read, aggregate and compare
Users	Operational staff	Managers and analysts
Example	Online order	Regional sales trend

Source systems → ETL/ELT → Staging → Data warehouse → Data marts / OLAP → Reports and decisions

9. Data Mining and Knowledge Discovery

9.1 Data Mining

Data mining is the discovery of useful patterns, relationships, trends and anomalies from large datasets using statistical, computational and machine-learning techniques.

9.2 KDD Process

Knowledge Discovery in Databases is the broader process of obtaining knowledge from data. A common sequence is selection, cleaning, integration, transformation, data mining, pattern evaluation and knowledge presentation.

9.3 Techniques

Technique	Meaning	Example
Classification	Assigns records to predefined categories.	Risk category
Regression	Predicts a numerical value.	Sales prediction
Clustering	Groups similar records without labels.	Customer segments
Association rules	Finds items that occur together.	Bread and butter
Anomaly detection	Identifies unusual behavior.	Suspicious transaction
Sequence analysis	Finds time-ordered patterns.	Steps before purchase

10. Important PYQ-Style Questions

1. What is the primary goal of BI? To convert data into useful insights that improve strategic, tactical and operational decisions.
2. What is GDSS? An interactive system supporting group communication, idea generation, evaluation, voting and collective decisions.
3. What are the components of GDSS? Hardware, software tools, people and network facilities.
4. What is groupware? Software enabling network-based communication, coordination and collaboration.
5. What is asynchronous groupware? Groupware supporting collaboration at different times, such as email, forums, workflow and file sharing.
6. What is DSS? An interactive system using data and models to support an individual decision-maker.
7. Differentiate DSS and GDSS. DSS supports an individual; GDSS supports a group and adds collaboration and voting.
8. What is OLAP? Multidimensional analysis of historical and integrated data.

11. Long-Answer Practice

- Define BI and explain its goals, components, process and applications.
- Explain how BI transforms raw data into meaningful insights.
- Define DSS and explain its components, types and characteristics.
- Differentiate Information System and DSS.
- Define GDSS and explain its components, phases, features, advantages and limitations.
- Differentiate DSS and GDSS with examples.
- Explain groupware and synchronous/asynchronous groupware.
- Explain OLTP, OLAP, data warehouses and data marts.
- Explain data mining and knowledge discovery techniques.
- Explain the decision-making process and models.

12. Quick Revision Sheet

Term	One-line recall
BI	Converts organizational data into decision insights.
DSS	Supports an individual using data and models.
GDSS	Supports group decisions with collaboration and voting.
Groupware	Network-based group communication and collaboration.
OLTP	Fast routine transactions and current data.
OLAP	Multidimensional analysis of historical data.
Data warehouse	Integrated historical repository for reporting.
Data mart	Smaller subject-specific warehouse section.
Data mining	Discovers patterns in large data.
KPI	Measurable indicator of performance against a goal.

13. Exam-Writing Strategy

For a 10-mark answer, write a definition, draw a process or architecture diagram, explain each component under separate headings, add advantages or applications, and conclude briefly. Use examples from retail, banking, healthcare, education or e-commerce.

14. Accuracy Notes

- BI includes collection, integration, storage and analysis—not only dashboards.
- DSS supports human judgment and does not replace managers.
- GDSS includes structured idea generation, evaluation, voting and documentation.
- OLTP is optimized for transactions, while OLAP is optimized for analysis.
- A data warehouse is designed for analysis, not routine transactions.
- Data mining is one stage of the broader KDD process.
- Asynchronous collaboration does not require simultaneous online presence.

15. Seven-Day Plan

Day	Target
1	BI meaning, goals, components and process
2	Analytics, tools, applications and advantages
3	Decision-making process and models
4	DSS definition, components, types and characteristics
5	GDSS definition, components, phases and evaluation
6	Groupware, OLTP, OLAP, warehouse, marts and data mining
7	Write long answers and attempt a timed mock test

Prepared from the supplied BCA-5001 Unit-I material and syllabus document. Language, structure and technical wording have been corrected and expanded for easy learning and university examination preparation.